Développement en série entière en 0	**Domaine de validité**
$e^x = \sum_{n=0}^{+\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	$x \in \mathbb{R}$
$\cos x = \sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$	$x \in \mathbb{R}$
$\sin x = \sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$	$x \in \mathbb{R}$
$\operatorname{ch} x = \sum_{n=0}^{+\infty} \frac{x^{2n}}{(2n)!} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$	$x \in \mathbb{R}$
$\operatorname{sh} x = \sum_{n=0}^{+\infty} \frac{x^{2n+1}}{(2n+1)!} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$	$x \in \mathbb{R}$
$\frac{1}{1-x} = \sum_{n=0}^{+\infty} x^n = 1 + x + x^2 + x^3 + \cdots$	$x \in]-1, 1[$
$\frac{1}{1+x} = \sum_{n=0}^{+\infty} (-1)^n x^n = 1 - x + x^2 - x^3 + \cdots$	$x \in]-1, 1[$
$\ln(1-x) = -\sum_{n=1}^{+\infty} \frac{x^n}{n} = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots$	$x \in]-1, 1[$
$\ln(1+x) = \sum_{n=1}^{+\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$	$x \in]-1, 1[$
$\arctan x = \sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$	$x \in [-1, 1]$
$(1+x)^\alpha = \sum_{n=0}^{+\infty} \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!} x^n = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2} x^2 + \ldots$	$x \in]-1, 1[$